MILITARY SPECIFICATION

TRAINING SET, CHEMICAL AGENT IDENTIFICATION, SIMULANTS, M72A1

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a simulant chemical agent identification training set which contains no chemical agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-676 - Boxes, Setup
PPP-C-843 - Cushioning Material, Cellulosic
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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MIL-P-116 - Preservation, Methods of
MIL-D-51256 - Detector Kit, Chemical Agent, ABC-M18A2
MIL-P-51409 - Paper, Chemical Agent Detector, VGH, ABC-M8; Booklet of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 5-77-2140 - Training Set, Chemical Agent Identification, Simulants, M72A1.

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM 3-6910-226-10 - Operator's Manual, Training Set, Chemical Agent Identification: Simulants, M72A1

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-77-2140 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

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3.1.2 Components. All components of the training set shall conform to the specifications and drawings listed on TDPL 5-77-2140 and subsidiary drawings.

3.2 Manufacture and assembly. The training set shall be manufactured and assembled in accordance with Drawing 5-77-2140 and subsidiary drawings.

3.3 Functioning. The simulant from each simulant bottle and from each ampoule shall produce the required color change when tested as specified in 4.4.4.1.

3.4 Ampoule leakage. Each type of ampoule shall show no evidence of leakage when tested as specified in 4.4.4.2.

3.5 Workmanship. The training set shall be free from foreign matter such as grease, oil, and dirt. The training set shall be free from visible damage such as cracks and dents.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of training sets shall be produced in accordance with this specification for examination and test (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

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4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five training sets shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production. The sample training sets shall be forwarded for tests to the test facility specified in the contract or purchase order (see 6.2).

4.3.2 Inspection procedures.

4.3.2.1 For examination. The five sample training sets shall be examined for the defects listed in 4.4.3.1.

4.3.2.2 For test. The five sample training sets shall be tested for functioning as specified in 4.4.4.1. Five of each type of ampoule shall be tested per set. All the ampoules in the training sets shall be tested for leakage as specified in 4.4.4.2.

4.3.3 Acceptance/rejection criteria. The presence of any examination defect in any of the preproduction samples or the failure of the functioning test or leakage test by any of the preproduction samples shall be cause for rejection of the preproduction lot.

4.4 Quality conformance inspection.

4.4.1 Lotting.

(a) Components. A lot shall consist of like components manufactured by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

(b) Training sets. A lot shall consist of the training sets manufactured by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of each type of simulant shall be represented in any one lot of finished training sets.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test.

(a) Functioning. Sampling for the functioning test shall be conducted in accordance with table I. Sample training sets shall be taken from among those samples examined in 4.4.3.1.

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TABLE I. Sampling for functioning test

Number of training sets in lot	Number of sample training sets to be tested
2 to 15	2
16 to 25	3
26 to 90	5
91 to 150	8
151 to 500	13
501 to 1,200	20
1,201 to 10,000	32
10,001 to 35,000	50
Over 35,000	80

(b) Ampoule leakage. Sampling for the leakage test for each type of ampoule shall be conducted separately in accordance with MIL-STD-105 using an AQL of 0.65 percent defective.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample training sets and the packaging thereof shall be examined in accordance with MIL-STD-105, inspection level S-4, and the following classification of defects:

(a) Training set, chemical agent identification, simulants, M72A1 (Drawing 5-77-2140).

Categories		Defects
Critical:		None defined
Major:		AQL 0.65 percent defective
	101	Component missing
	102	Component incorrectly assembled
	103	Workmanship (see 3.5)

(b) Test bottle assembly (Drawing 5-77-2141).

Categories		Defects
Critical:		None defined
Major:		AQL 2.5 percent defective
	101	Bottle dimensions not as specified
	102	Marking on bottle or cap incorrect, missing, or illegible

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(c) Simulant bottle assembly (Drawing 5-77-2148).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Bottle dimensions not as specified
102	Contents missing
103	Cap not securely tightened
104	Marking on bottle or cap incorrect, missing, or illegible

(d) Vial box carrier (Drawing 5-77-2191).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dimensions incorrect

(e) Ampoule assembly (Drawing 5-77-2143).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Length incorrect
102	Fill volume incorrect

(f) Simulant and test bottle carrier (Drawing 5-77-2357).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Dimensions incorrect

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(g) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Contents of vial boxes, carriers, or pail missing or not as specified
102	Vials of ampoules in paperboard box not packaged as specified
103	Arrangement of empty test bottles and filled simulant bottles in carrier not as specified
104	Packing not as specified
105	Pail or pail closure not as specified or damaged
106	Fiberboard pads missing or not as specified
107	Cushioning material missing or not as specified (where needed)
108	Marking of vial boxes incorrect, missing, or illegible
109	Marking of pail incorrect, missing, or illegible

4.4.3.2 For test.

(a) Functioning. The components taken as specified in 4.4.2.2(a) shall be tested as specified in 4.4.4.1. Failure of any component to produce the required color change shall be cause for rejection of the lot represented.

(b) Ampoule leakage. The sample ampoules taken as specified in 4.4.2.2(b) shall be tested as specified in 4.4.4.2.

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning.

(a) Test bottles and ampoules. Using the necessary components from an M18A2 detector kit conforming to MIL-D-51256, sample and test the vapor from each type of test bottle in the training set and observe for the required color change in accordance with the applicable instructions provided with the M18A2 detector kit and the instructions in TM 3-6910-266-10.

(b) Simulant bottles. Test each type of simulant bottle in the training set as follows: Using the applicator in the vial cap, place one drop of simulant on M8 detector paper conforming to MIL-P-51409 and observe for the required color change in accordance with the instructions provided with the M8 detector paper and the instructions in TM 3-6910-226-10. Replace the cap.

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4.4.4.2 Ampoule leakage. Place the ampoules under colored water in a desiccator using a suitable cover to keep the ampoules under water during the test. Reduce the pressure in the desiccator to 5 inches of mercury or less absolute and maintain at that pressure for at least 25 minutes, and then pressurize to ambient. After no less than 20 minutes at ambient pressure, remove the ampoules from the water, rinse thoroughly in clean water, and spread on a clean surface. Colored water in an ampoule shall be considered evidence of leakage.

5. PACKAGING

5.1 Unit packing, level A. Each of the nine types of ampoules shall be unit packed as follows: Fifty vials containing the same type of ampoules shall be placed with the closure end upright in a setup paperboard box conforming to type I, variety 1, class C, style 4 of PPP-B-676. The inside dimensions of the box shall be 101.6 millimeters (mm) by 47.6 mm by 50.8 mm for length, width, and height, respectively. Fiberboard pads shall be inserted flat against the inner faces of the ends of the box in sufficient quantity to prevent movement of the vials. The pads shall be formed from material conforming to type CF, class domestic, variety and grade optional of PPP-F-320. Fiberboard pads shall be 47.6 (+0 or -1.6) mm wide. The height of the pads shall be $\pm$ 1.6 mm of the length of the vial shell including the cap (Drawing 5-77-2145 and 5-77-2142).

5.2 Packing, level A. Three filled molded polystyrene carriers shall be placed in a pail as shown on Drawing 5-77-2137. The bottom carrier (Drawing 5-77-2140) shall contain five boxes of vials unit packed as specified in 5.1. The middle carrier (Drawing 5-77-2140) shall contain four boxes of vials, unit packed as specified in 5.1. All boxes of vials shall be packed so that the marked end of the box is visible. The top carrier (Drawing 5-77-2140) shall contain three filled simulant bottles (Drawing 5-77-2148) and eight empty test bottles (Drawing 5-77-2141). Cushioning material conforming to cellulosic, type I, class A of PPP-C-843 shall be placed in the carriers as needed to prevent movement of contents and to ensure a tight fit. A sufficient quantity of fiberboard conforming to type CF, class domestic, grade 125 of PPP-F-320 shall be placed on top of the top carrier to ensure a tight fit. A copy of the technical manual for the training set shall be placed on top of the fiberboard and the other contents of the pail.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. The top of the cover and both ends of each box of vials, unit packed as specified in 5.1, shall be marked for identification with black printer's ink by showing the symbol (minimum 6.35 mm letters) corresponding to the identification marking on the contents. The pail shall be marked as shown on Drawing 5-77-2363.

5.3.1 Precautionary marking. The top cover of each box of vials shall be marked as follows:

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WARNING!

Harmful if inhaled or swallowed.
Avoid contact with eyes, skin, and clothing.
Use with adequate ventilation.
Wash thoroughly after handling.

6. NOTES

6.1 Intended use. The training set covered by this specification is intended for use with the M18A2 Chemical Agent Detector Kit in training personnel in the identification of certain chemical agents.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification, and
(b) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract,
(2) Name and address of test facility and shipping instruction when testing is performed by the Government, and
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Shipping permit. Refer to special permit DOT-E-6731 for shipping regulations.

6.4 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-T-1095, dated 25 July 1972.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 6910-A432

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-51497(EA)	TRAINING SET, CHEMICAL AGENT IDENTIFICATION, SIMULANTS, M72A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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